

# Phenotype Concept Recognition

## *Workflow, terminology, error taxonomy, and methodological approaches*

 **What** Automated phenotype extraction from clinical notes. Performed by open source tools such as: ClinPhen, Phenotagger, HPO-RAG, PhenoBCBert, etc.

 **Why** Deep phenotyping is essential to the rare disease diagnostic process, requiring extensive clinical subspecialty expertise. Much of this time-consuming work can be automated, alleviating the burden to clinical staff and surfacing important information otherwise buried in patient clinical notes.

 **How** This is accomplished by a process called **Phenotype Concept Recognition** which 1) recognize phenotypes in the unstructured text of clinical notes and 2) assigns them to standardized terms from ontologies such as HPO

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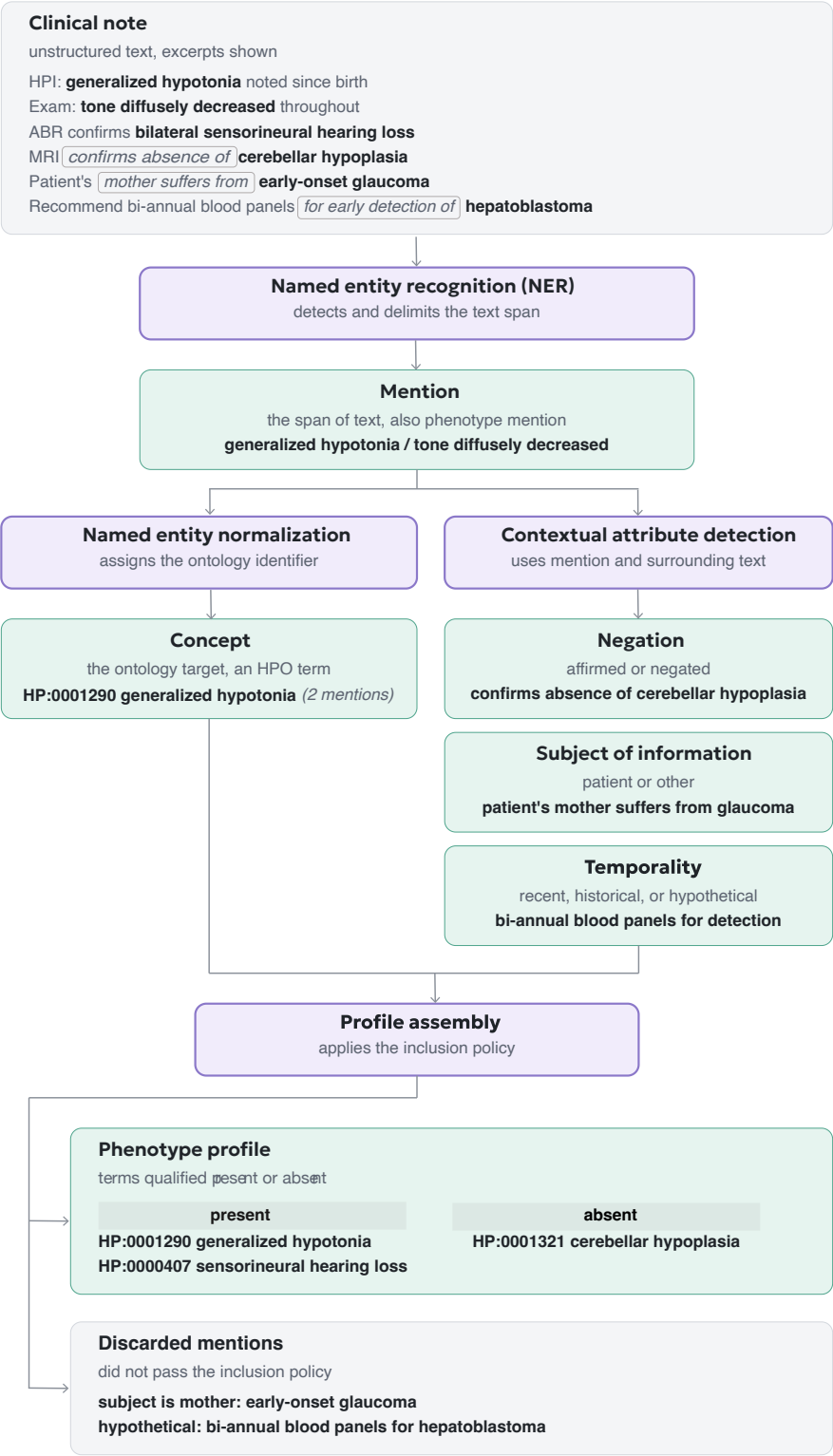
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# 1. Overview - Key Tasks and Data Objects

Here, the term **Phenotype Concept Recognition** is used to describe the computational tools, such as ClinPhen, that generate candidate HPO terms from clinical notes, taking unstructured text as input to identify **phenotype mentions** (text spans) that are then linked to ontology **concepts** (e.g. HPO terms). Two key tasks are performed: **Named Entity Recognition**, (aka, Entity linking, or concept mapping/assignment mapping to ontology which identifies the text block that describes a phenotype, and Named entity normalization, also called entity linking or ontology-based concept

Phenotype Concept Recognition  
Clinical NLP Lexicon



## 2. Dictionary of terms

Supplementary reference for Pheno+. Terminology follows clinical NLP convention (i2b2/n2c2, BioCreative) where one exists, and is flagged explicitly where it does not.

### 2.1 Foundational

#### **Concept recognition (CR)**

The foundational task. Locating ontology-relevant entities in text and linking them to ontology identifiers — that is, named entity recognition plus named entity normalization performed together. Also written *ontology-based concept recognition*. All other task terms in this document are subtasks or neighbors of CR.

#### **Phenotype concept recognition (PCR)**

Concept recognition specialized to phenotype descriptions and an ontology of phenotypic abnormality, in practice the HPO.

#### **Phenotype extraction**

The applied-genomics phrase for the same activity. Acceptable in a title or abstract aimed at a genomics readership; not a technical NLP task name. Define once as concept recognition, then use CR vocabulary in methods.

#### **Deep phenotyping**

Comprehensive, ontology-encoded phenotypic characterization of an individual. A clinical-genetics framing term for the introduction, not a methods term.

### 2.2 Tasks

#### **Named entity recognition (NER)**

Detects and delimits the character span expressing a concept. Does not assign an identifier. Synonyms: *mention detection*, *span detection*, *entity boundary detection*. The label "named entity" is a legacy inheritance from newswire NER; biomedical entities are common nouns, not proper names. Use the phrase only inside the fixed task names.

#### **Named entity normalization (NEN)**

Links a mention to a concept in a standardized terminology. In clinical NLP the more idiomatic running form is *concept normalization*; the 2019 n2c2/UMass Lowell shared task on this problem is titled clinical concept normalization, and the task paper uses *medical concept normalization (MCN)*. The recent neural literature (BioSyn, SapBERT and successors) calls the same task *biomedical entity linking*; note the equivalence once on first use.

#### **Contextual attribute detection**

Assigns contextual properties to a recognized concept. Executed as three independent axes,

each with a default value. Also *attribute detection*, *context classification*.

### Profile assembly

Applies an inclusion policy to a concept and its axis values, determining whether it is admitted to the phenotype profile and with what qualification. **Not a term of art** — no settled NLP name exists because most tools bury this in post-processing and never describe it. Define on first use. Alternatives: *inclusion policy*, *profile filtering*.

### Section detection

Segmenting a note into named sections (History of Present Illness, Family History, Physical Exam). Also *section segmentation*. A preprocessing stage; much family-history leakage is a section-detection failure rather than a matcher failure.

## 2.3 Text-side objects

### Clinical note

The authored artifact a clinician produces during or after an encounter. Use for the input artifact and when describing clinician workflow.

### Clinical free text / unstructured text

The prose itself, as opposed to coded fields. Use when the contrast is with discrete EHR data. Related: *clinical narrative*, *clinical text*. Caution: FHIR uses Narrative for the human-readable XHTML rendering of any resource, so *clinical narrative* is ambiguous in FHIR-facing sections.

### Document

The processing and evaluation unit. Correct in "document-level evaluation," "per-document F1," "the corpus contains 500 documents."

### Note type

The categorical variable: discharge summary, progress note, history and physical, consultation note, genetics evaluation. Report it — performance varies substantially across types, and most tools were evaluated on ED and discharge-summary corpora.

### Corpus

The collection. Specify document count, note-type distribution, and de-identification status.

### Mention

An occurrence of a concept in text — the unit NER outputs and NEN consumes. The shared tasks legislate this usage. *Phenotype mention* on first use, then *mention*. *Concept mention* is an equivalent compound.

### Span

The character offsets. Use when you mean coordinates rather than the linguistic unit.

### Surface form

The particular lexical realization in text, as opposed to the ontology's preferred label. "Floppy

baby" and "hypotonia" are different surface forms of one concept.

### Annotation

The record pairing a span with a concept identifier. The unit in a reference standard.

### Trigger term / cue term

A lexical item that causes a contextual attribute to depart from its default value.

### Scope

The region of text over which a trigger's value applies, extending forward until the sentence ends or a termination term fires.

### Termination term

A lexical item that closes a trigger's scope. Without termination handling, one negation cue suppresses everything downstream to the period.

## 2.4 Ontology-side objects

### Human Phenotype Ontology (HPO)

Specify the release version in methods. Term counts, labels, and synonyms change between releases.

### Concept

The abstract task-level noun and the target of normalization. Entrenched: UMLS identifiers are Concept Unique Identifiers. Needs no *ontology concept* qualifier past first use.

### Term

HPO's own register — the HPO literature says "HPO term," not "HPO concept." Use when naming a specific ontology target.

### Class

The formal OWL construct. Reserve for structural discussion: ancestor classes, `is_a` edges, information content.

### Primary label

The ontology's canonical name for a term.

### Synonym

An alternate label registered in the ontology, typed by *scope* (exact, broad, narrow, related — the four `obol:Owl` predicates) and optionally by *type* (e.g. `layperson`). Together with the primary label, synonyms constitute the lexicon available to dictionary-based matching, and therefore bound its recall. Roughly 36% of HPO terms carry at least one `layperson` synonym; a substantial fraction carry no synonym at all. **State which scopes you loaded** — including broad synonyms systematically induces over-specification errors, and almost no paper reports this.

### CURIE

Compact URI, `prefix:reference` — `HP:0001250`. W3C-specified, resolvable via a prefix registry

(OBO Foundry, identifiers.org, Bioregistry). This is **semantic-web and OBO vocabulary, not clinical informatics vocabulary**. Clinical informatics uses a `system` + `code` pair. Use CURIE in genomics-facing sections and in any Phenopacket discussion; use *HPO term identifier* for a clinical informatics venue. The prefix is `HP:`, not `HPO:`. Avoid "HPO code" entirely.

### Phenotype profile / phenotypic profile

The set of HPO terms characterizing one individual. The HPO and Phenopacket literature more often uses *phenotypic profile*; match the figure to the running text.

### Present / absent

The qualification carried by a term admitted to the profile. A negated finding in the patient is a positive diagnostic result and should be retained as absent, not discarded — Exomiser and LIRICAL both consume excluded features.

### Granularity / specificity

Depth in the `is_a` hierarchy. Linking "seizure" to "Abnormality of the nervous system" is under-specification; linking to a child term the note does not support is over-specification.

## 2.5 Contextual axes

### Negation

Whether the concept is affirmed or negated. Default: affirmed. Qualifies the term and retains it.

### Subject of information

Whom the finding is attributed to. Default: patient. Equivalent to ConText's *experiencer* axis and to the i2b2 label *associated with someone else*. Gates the term. HL7 clinical-information-modeling vocabulary, contrasting with *subject of record*.

### Temporality

Whether the finding is recent, historical, or hypothetical. Default: recent. Empirically the weakest of the three axes in every published ConText evaluation.

### Assertion status

The alternative i2b2/n2c2 2010 convention: a single flat label per mention drawn from present, absent, possible, conditional, hypothetical, associated with someone else. Mutually exclusive, so a negated family-history mention cannot be represented on two axes. Choose the ConText orthogonal-axis convention or this one and hold it — the choice determines whether your error taxonomy can separate negation from subject errors.

## 2.6 FHIR and EHR integration

### SMART on FHIR

The app-integration specification governing launch and authorization.

### EHR launch / standalone launch

Whether the app is launched from within the EHR with patient context supplied, or independently.

Reviewers will look for the distinction.

### **Launch context**

The contextual parameters (patient, encounter, user) passed at launch.

### **Scopes**

OAuth 2.0 authorization strings governing resource access. State them explicitly (e.g. `patient/DocumentReference.read`) — this is what makes an implementation paper reproducible.

### **DocumentReference**

Metadata *about* a clinical document, not the document. The text lives in

`DocumentReference.content.attachment`, either inline base64 or a `url` pointing to a `Binary`.

"Retrieved via DocumentReference" is accurate; the pipeline consumes the attachment.

### **DocumentReference.type**

The note type, usually carried as a LOINC document-type code.

### **Binary**

The FHIR resource holding raw document bytes when not inlined in the attachment.

### **Observation**

A candidate write-back target for a phenotype finding.

### **Observation.subject**

Whose record the observation appears in — always the patient, even when the finding concerns someone else. Do **not** map a non-patient subject of information here.

### **Observation.focus**

The person or entity a finding is about when that differs from the subject. This is where a non-patient subject of information maps, if retained at all.

### **Condition**

Alternative write-back target. Observation vs. Condition is a real modeling decision reviewers will ask about.

### **FamilyMemberHistory**

The FHIR resource for a relative's conditions. The structurally correct destination for family-history findings.

### **Write-back**

Persisting reviewed results into the EHR as discrete data.

### **Discrete data**

The clinical informatics term for coded, queryable fields. Preferable to the vaguer "structured."

### **US Core**

The US FHIR implementation guide profiling the resources above. Cite the version you conform to.

## 2.7 GA4GH Phenopackets

### Phenopacket Schema

The GA4GH standard for sharing disease and phenotype information characterizing an individual or biosample. ***Carries more context than a bare HPO term list*** — *onset age, severity, resolution, and HPO modifiers*. The natural output contract for a phenotype extraction app. Published as ISO 4454:2022.

### PhenotypicFeature

The schema element representing one phenotype observation, qualified as ***present or excluded***, with optional ***severity, modifiers, onset, and resolution***.

### OntologyClass

The schema element holding an ontology term as a CURIE plus label. This is why CURIE vocabulary is appropriate in Phenopacket-facing sections.

### excluded (not to be confused with "discarded")

The Phenopacket qualifier for a ***confirmed-absent*** feature. Note: in the Phenopacket lexicon ***excluded means confirmed absent, it DOES NOT mean "filtered out" or "discarded"***. A phenopackets entry is, by default, presumed to be confirmed as present in the patient. To indicate that a phenotype is NOT present in the patient, the "excluded" qualifier is used. Use *discarded* for mentions that fail the inclusion policy.

### Phenopackets on FHIR

A FHIR Implementation Guide based on the Phenopacket standard, described by GA4GH as conceptually equivalent but **not directly interoperable**, and under further development within the HL7 FHIR Vulcan program. Quote that caveat if you claim Phenopacket compatibility.

## 2.8 Evaluation and human-in-the-loop

### Reference standard / gold standard

Manually annotated ground truth. *Reference standard* is preferred in clinical informatics venues, *gold standard* in NLP.

### Silver standard

Automatically or weakly generated annotations. Always flag as such.

### Inter-annotator agreement (IAA)

Report with the statistic named and the number of annotators.

### Exact match / relaxed match

Whether span boundaries must coincide exactly or merely overlap. The single largest source of incomparable F1 values across phenotype extraction papers. Report both.

### Mention-level / concept-level / document-level evaluation

Whether each occurrence, each unique concept, or the per-document term set is scored.

Document-level is usually what matters clinically. Almost no paper says which it used; saying so is a cheap credibility win.

### Human-in-the-loop (HITL)

The design pattern in which **a clinician reviews automated output** before it is committed. In Pheno+, the clinician plays an active role in assembling the phenotypic profile by 1) choosing from the patient's chart, the notes that cover and characterize the patient's phenotypes 2) reviewing the generated HPO terms, by accepting the terms to include and manually adding any missing terms. This serves as both quality control and curation.

### Clinician review

The routine accept / reject / edit workflow.

### Adjudication

Resolving disagreement — between annotators, or between tools. Do not use interchangeably with review.

### Provenance

The link from an accepted term back to its source span and document. The central affordance of a review interface.

### Review burden / curation burden

Clinician time and effort required.

### Acceptance rate

Proportion of suggested terms accepted. Define the denominator explicitly if edits and rejections are tracked separately.

## 3. Error taxonomy

Errors are assigned to the **earliest task at which they could have been prevented**. Stating this precedence rule up front makes the taxonomy a partition and keeps counts additive; without it, boundary errors that produce bad links get double-counted and the categories will not sum.

| Preferred term | Description                                                                                                                                                                       | Task |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| Missed mention | Phenotype described in text but no span produced. Sub-causes: surface form absent from the lexicon, descriptive rather than nominal phrasing, abbreviation, misspelling.          | NER  |
| Boundary error | Span found with wrong extent — over-extension pulls in modifiers or anatomy, truncation drops a qualifying head noun. Drives divergence between exact-match and relaxed-match F1. | NER  |

| Preferred term                   | Description                                                                                                                                                                            | Task                                      |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| Discontinuous mention failure    | Span interrupted by intervening tokens ("long fingers and toes"). Typically yields one correct term and one miss.                                                                      | NER                                       |
| Nested mention failure           | A shorter valid term inside a longer one ("cardiomyopathy" within "hypertrophic cardiomyopathy"). Systems assuming non-overlapping spans cannot recover both.                          | NER                                       |
| Spurious span                    | Non-phenotype text tagged. Common sources: medication names, procedure names, bare anatomical terms, devices.                                                                          | NER                                       |
| Coordination / ellipsis failure  | Shared head or modifier not distributed ("left and right ptosis", "mild hypotonia and hyporeflexia").                                                                                  | NER                                       |
| Candidate generation failure     | Correct term never enters the candidate set. Root cause is almost always synonym coverage.                                                                                             | NEN                                       |
| Disambiguation error             | Correct candidate retrieved, wrong one selected. Worst case is abbreviations (PDA, MR).                                                                                                | NEN                                       |
| Under-specification              | Linked to an ancestor of the correct term.                                                                                                                                             | NEN                                       |
| Over-specification               | Linked to a descendant the text does not support. Systematically induced by loading broad synonyms into the lexicon.                                                                   | NEN                                       |
| Sub-ontology error               | Linked into Mode of inheritance, Clinical course, Clinical modifier, or Frequency instead of Phenotypic abnormality.                                                                   | NEN                                       |
| Forced link (NIL failure)        | The finding has no adequate ontology representation and the system links anyway rather than abstaining. <i>NIL prediction</i> is correctly declining to link.                          | NEN                                       |
| Modifier loss                    | Severity, laterality, or onset qualifiers dropped during linking. Not an ontology error, but a loss of information the Phenopacket schema can represent.                               | NEN                                       |
| Negation leakage                 | Negated finding recorded as present. Causes: missing trigger, concept outside scope, or scope over-run from missing termination term.                                                  | Contextual attribute detection — negation |
| Over-negation                    | Affirmed finding suppressed because scope crossed a clause boundary that should have terminated it.                                                                                    | Contextual attribute detection — negation |
| Uncertainty conflation           | Hedged findings ("cannot exclude seizures", "no definite hypotonia") forced into a binary affirmed/negated decision.                                                                   | Contextual attribute detection — negation |
| Family-history attribution error | A relative's phenotype enters the proband profile. Separate cue-level causes (relation term present but unrecognized) from section-level causes (Family History section unrecognized). | Contextual attribute detection — subject  |

| Preferred term                  | Description                                                                                                                                                                 | Task                                         |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| Over-attribution to non-patient | The proband's own finding excluded because it sat near a relation term. Invisible unless the reference standard annotates discarded mentions.                               | Contextual attribute detection — subject     |
| Non-family non-patient subject  | Fetus, twin, donor, or unspecified third party. Fails differently from family history; a binary patient/other value set collapses these.                                    | Contextual attribute detection — subject     |
| Historical-as-current           | Resolved or remote findings admitted as active.                                                                                                                             | Contextual attribute detection — temporality |
| Hypothetical-as-present         | Conditional and anticipatory framings admitted ("if she develops seizures", "monitor for hypotonia").                                                                       | Contextual attribute detection — temporality |
| Surveillance-as-finding         | Planned procedures, ordered tests, and pending results treated as observed findings.                                                                                        | Contextual attribute detection — temporality |
| Cascading error                 | Upstream span error corrupting downstream linking. Exists only in pipeline architectures; naming it allows an evidence-based argument for or against joint modeling.        | Architecture                                 |
| Attribute scope mismatch        | Attributes resolved at sentence level while the governing context sits in an adjacent sentence or a section header.                                                         | Architecture                                 |
| Redundancy                      | Both an ancestor and its descendant admitted. Not wrong, but distorts downstream semantic-similarity scoring in Exomiser or LIRICAL.                                        | Profile assembly                             |
| Deduplication failure           | The same concept from multiple mentions counted repeatedly, inflating document-level metrics.                                                                               | Profile assembly                             |
| Intra-document conflict         | The same concept affirmed in one sentence and negated in another, with no resolution policy.                                                                                | Profile assembly                             |
| Ontology version drift          | Obsolete or merged terms persisting in a stored profile across HPO releases. Systemic rather than per-mention; affects reproducibility of anything written back to the EHR. | Profile assembly                             |

## 4. Approaches to phenotype concept recognition

| Approach                       | Mechanism                                                | Strengths                                                  | Limitations                                                | Representative open-source tools                       |
|--------------------------------|----------------------------------------------------------|------------------------------------------------------------|------------------------------------------------------------|--------------------------------------------------------|
| <b>Dictionary / rule-based</b> | String and pattern matching against a lexicon built from | <b><i>Fast, fully deterministic, no training data,</i></b> | <b><i>Recall bounded by synonym coverage; fails on</i></b> | <b>ClinPhen;</b><br>Doc2HPO;<br><b>FastHPOCR;</b> NCBO |

| Approach                                         | Mechanism                                                                                                                                               | Strengths                                                                                                                                                                                                  | Limitations                                                                                                                                                                                                      | Representative open-source tools                                                                                                                                                                                                                            |
|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                  | ontology primary labels and synonyms, with hand-written rules for negation and normalization.                                                           | <i>inspectable lexicon, trivially auditable, runs on commodity hardware</i> inside an enclave. Errors are diagnosable by reading the lexicon.                                                              | <i>descriptive and non-nominal phrasing</i> , discontinuous mentions, <i>abbreviations, misspellings</i> . Context handling only as good as the cue lexicon. <i>Mapping failures dominate the error profile.</i> | Annotator; cTAKES; MetaMap                                                                                                                                                                                                                                  |
| <b>BERT-based (encoder, span classification)</b> | A pretrained biomedical/clinical encoder fine-tuned for span labeling and concept assignment, sometimes with a CNN or hierarchical classification head. | Generalizes beyond the lexicon to unseen surface forms; <i>handles morphological and syntactic variation; can learn assertion</i> jointly with the span. <i>Modest inference cost, deployable locally.</i> | Requires annotated training data; performance degrades off the training note type; span-level architectures typically assume non-overlapping mentions; <i>the decision is not inspectable.</i>                   | <b>PhenoBERT</b> (two-level CNN over deep-learning representations); <b>PhenoBCBERT</b> (Bio+Clinical BERT); PhenoBrain phenotype extraction module (TopWords segmentation followed by ALBERT mapping to HPO terms); BioSyn / SapBERT for the linking stage |
| <b>Hybrid dictionary + neural</b>                | Dictionary matching supplies high-precision candidates and weakly-labeled training data; a BERT model extends coverage to unmatched spans.              | <i>Combines dictionary precision with neural recall</i> ; the weak-supervision route avoids the need for large manual annotation. <i>Currently among the most reliable published HPO recognizers.</i>      | Inherits both error profiles; still requires context handling as a separate stage, where <i>negation leakage and non-assertive context remain prominent</i> . Two components to tune and version.                | <b>PhenoTagger</b> (dictionary-based and BERT-based components combined)                                                                                                                                                                                    |

| Approach                                    | Mechanism                                                                                                                                                                                                                       | Strengths                                                                                                                                                                                                                          | Limitations                                                                                                                                                                                                                                      | Representative open-source tools                                                                                                                                                                                               |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Generative LLM (GPT-family)</b>          | An instruction-tuned or fine-tuned decoder model is prompted to emit phenotype terms, optionally with identifiers, directly from the note.                                                                                      | <b>Strong contextual sensitivity</b> — <b>handles negation, attribution, and hedging</b> that defeat cue lexicons. Can surface phenotype concepts not represented in the current ontology. Little or no task-specific engineering. | <b>Prone to hallucinated identifiers</b> ; unreliable at emitting valid CURIEs without grounding. Non-deterministic. Closed-weight models are typically unusable in a HIPAA enclave; <b>open-weight models require substantial GPU capacity.</b> | <b>PhenoGPT</b> (initializable from GPT-J, Falcon, LLaMA, or GPT-3/3.5)                                                                                                                                                        |
| <b>Retrieval-augmented generation (RAG)</b> | The LLM extracts candidate phenotypic phrases; each is matched by semantic similarity against a vector database of ontology-derived phrases; retrieved candidates are returned to the LLM as context for final term assignment. | Grounds the linking step in real ontology entries, <b>sharply reducing hallucinated identifiers while retaining LLM context sensitivity.</b> <b>Avoids fine-tuning cost.</b> Vector database is updatable without retraining.      | <b>Recall still bounded by the phrase database</b> ; retrieval quality depends on the embedding model; two-stage design reintroduces cascading error. <b>Inference cost higher than dictionary methods.</b>                                      | <b>RAG-HPO</b> — vector database of >54,000 phenotypic phrases mapped to HPO IDs; reported mean precision 0.84, recall 0.78, F1 0.80 with Llama-3 70B on a 120-case-report benchmark, against Doc2HPO, ClinPhen, and FastHPOCR |
| <b>Agentic</b>                              | An LLM orchestrates tool calls — ontology lookup, graph traversal, section retrieval, targeted re-reading — and reasons over intermediate results, optionally revising earlier decisions.                                       | Can perform explicit ontology traversal to resolve granularity; can seek additional evidence rather than committing on first pass; naturally accommodates a human-in-the-loop checkpoint. Attribution and                          | <b>No established open-source tool or published benchmark for HPO concept recognition specifically.</b> High latency and cost; error attribution is difficult across a multi-step trajectory;                                                    | No mature dedicated tool; built ad hoc on agent frameworks (LangGraph, PydanticAI) with ontology and knowledge-graph tool calls                                                                                                |

| Approach | Mechanism | Strengths                                                      | Limitations                                                                                                           | Representative open-source tools |
|----------|-----------|----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|----------------------------------|
|          |           | temporality can be reasoned about rather than pattern-matched. | reproducibility is poor without trajectory logging. Treat as an open research direction, not an evaluated comparator. |                                  |

Notes on the comparison

The published error profiles differ by approach in a way that matters for tool selection. *Dictionary methods fail predominantly at mapping* — the correct term exists but the surface form is not in the lexicon. *Neural and hybrid methods spread their errors across non-assertive context, negation leakage, and mapping.* If clinician review is the safety net, the two failure modes are not equally costly: *a mapping failure is a silent omission the reviewer cannot see, whereas a negation leak is a visible false positive the reviewer can reject.*

**Benchmarks are not comparable across papers** unless exact-versus-relaxed matching and mention-versus-document-level scoring are stated. Reported F1 values differ by more than the between-method gap depending on these choices.

**Deployment constraints may dominate accuracy.** Within a HIPAA enclave, closed-weight API models are typically unavailable, which removes several published comparators from consideration regardless of reported performance.

**PhenoBrain enters the table only as its extraction module.** PhenoBrain is a two-module diagnostic pipeline: a phenotype extraction module followed by an ensemble differential-diagnosis module that ranks candidate rare diseases. Only the first module is a concept recognizer; the second belongs with Exomiser and LIRICAL as a downstream consumer. It was developed and validated principally on Chinese-language EHRs, so extraction performance does not transfer to English clinical notes without revalidation.

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Generated by Claude Opus 5, Revised by Tony Di Sera

Terminology follows clinical NLP convention, drawing from interoperability standards (i2b2/n2c2, BioCreative, FHIR)